
Myco-Opt: Fungal-Network Priors for LLM-Driven Hyperparameter Optimization in Agricultural Earth Observation

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Abstract Hyperparameter optimization (HPO) remains sample-inefficient for Earth Observation (EO) machine-learning pipelines, where each evaluation can involve training on large multi-spectral time series. Recently, large language models (LLMs) have been proposed as optimizers (e.g., OPRO), but an unconstrained LLM proposer explores high-dimensional search spaces poorly when the evaluation budget is small. We introduce *Myco-Opt*, an LLM-driven HPO framework whose search space is constrained by *biological priors* extracted from a real ectomycorrhizal network (Beiler et al., 2010/2015): the same network topology that evolution shaped to transport water and carbon under uncertainty is used to shape where an LLM optimizer may search. We evaluate Myco-Opt on an agricultural EO regression task — forecasting daily reference evapotranspiration (ET_0) for avocado orchards in La Ligua, Chile, from 10 years of open satellite and reanalysis data. At a budget of 12 evaluations, the prior improves the LLM optimizer from RMSE 0.627 (OPRO-vanilla) to 0.614, and Myco-Opt’s *first* proposal (RMSE 0.616) already beats the best configuration the unconstrained LLM finds in 12 trials (0.621): quantitative biological topology provides a cheap, transferable prior that makes LLM-based HPO viable in the low-budget regime that dominates EO practice. All code, data and mappings are open source.

1 Introduction

Earth Observation (EO) is a demanding domain for automated ML: modern pipelines ingest multi-spectral satellite archives, foundation-model embeddings and dense reanalysis (Tuia et al., 2021; Ruwurm et al., 2023). Finding good hyperparameters is wasteful when *each* evaluation requires training over long time series; in operational EO settings the budget is often in the tens, not hundreds, of trials.

Bayesian optimization (BO, e.g. TPE (Bergstra et al., 2011)) needs many evaluations to build a reliable surrogate. An alternative casts an LLM as the optimizer: OPRO (Yang et al., 2024) prompts an LLM with evaluation history to propose the next configuration — no surrogate needed. But when the space is wide and the budget small, an unguided LLM explores poorly: its first proposals are no better than random.

In this work we ask: *can a quantitative prior derived from real biological network data make an LLM optimizer competitive in the low-budget regime?* We answer affirmatively with **Myco-Opt**, whose search space is derived from the topology of an ectomycorrhizal network — the underground “wood-wide web” plants use to transport water, carbon and nutrients under uncertainty (Simard et al., 1997; Beiler et al., 2010). These networks are evolved solutions to resource-flow optimization under stress; their measured topology provides a natural prior over the hyperparameters of ML pipelines that must operate under hydrological stress.

We evaluate Myco-Opt for agricultural EO water management: Chile, the world’s third-largest avocado exporter, faces a structural drought in its central zone, where 37,000 ha of orchards consume 9,000 m³/ha/year (CAZALAC-UNESCO). Precision irrigation requires forecasting reference evapotranspiration (ET_0). We forecast daily ET_0 one day ahead for La Ligua (center of the

Metric	Symbol	Measured value
Modularity (Louvain)	Q	0.5375
Scale-free exponent	γ	1.059
Small-world coefficient	σ	2.376
Average path length	$\bar{L}$	1.584
Average clustering	C	0.092

Table 1: Network metrics of the real mycorrhizal common-network graph (Beiler et al. 2010/2015, 67 trees, 144 CMN edges). These numbers are the biological prior used to constrain the LLM’s search space in Myco-Opt.

Chilean avocado belt) from 10 years (2016–2026) of open satellite/reanalysis data. Our contributions: (i) **Myco-Opt**, an LLM-driven HPO method whose search space is constrained by metrics measured on a real, published mycorrhizal network rather than by heuristics; (ii) an **agricultural EO benchmark** (daily ET_0 forecasting for avocado orchards from open EO data, strict temporal splits, full reproducibility); (iii) an **empirical ablation** showing that at budget 12 the prior improves the identical LLM optimizer (OPRO-vanilla) substantially, while classical TPE remains the strongest baseline at this budget (Section 5).

2 The biological prior

2.1 Mycorrhizal networks as optimized transport systems

Ectomycorrhizal fungi form physical networks connecting the roots of multiple plants, exchanging water, carbon and nutrients (Simard et al., 1997). Shaped by evolution under drought and resource heterogeneity, common-mycorrhizal networks (CMNs) empirically exhibit *scale-free* degree distributions (few highly connected “hub” trees), *modular* community structure (guilds specialized on subsets of hosts), and *small-world* connectivity (Beiler et al., 2010; Southworth et al., 2014).

Myco-Opt’s working hypothesis is modest and falsifiable: **these structural features, measured on a real network, define a good region of the hyperparameter space for ML models that must generalize under environmental noise**. We claim no fungal “intelligence”; we claim only that quantitative network topology provides a useful, transferable prior.

2.2 Data: the Beiler Douglas-fir CMN

To keep the prior *empirical*, we use the canonical dataset of Beiler et al. (2010, 2015): a 30×30 m plot of interior Douglas-fir in British Columbia, where 67 trees are connected by 27 genet of two *Rhizopogon* species mapped by molecular identification of shared genets. From the tree-to-tree common-mycorrhizal graph (67 nodes, 144 edges, edge weight = number of shared genets) we compute five NetworkX (Hagberg et al., 2008) metrics:

2.3 Mapping topology to hyperparameters

Each metric maps monotonically and a priori to a LightGBM hyperparameter (Ke et al., 2017) (Table 2): high modularity (specialized guilds) $\Rightarrow$ more leaves (specialized partitions); short average path $\Rightarrow$ shallow trees. The prior does *not* fix values; it only *restricts the search range* around a topology-derived center, leaving the LLM free to explore and the evaluation to decide.

3 The EO benchmark task

3.1 Domain: precision irrigation for avocado under drought

Avocado (Hass) is central Chile’s highest-value export crop and one of its most water-intensive ($\sim 9,000 \text{ m}^3/\text{ha}/\text{year}$). Irrigation decisions require knowing tomorrow’s atmospheric water demand:

Fungal metric	Biological reading	LightGBM parameter (range)
$Q = 0.5375$	specialized guilds	<code>num_leaves</code> $\in [42, 66]$
$\tilde{L} = 1.58$	shallow transport	<code>max_depth</code> $\in [3, 7]$
$C = 0.092$	low local redundancy	<code>reg_lambda</code> $\in [0.1, 3.1]$
$\gamma = 1.06$	near-homogeneous degree	<code>colsample</code> $\in [0.64, 0.84]$
$\sigma = 2.38$	small-world shortcuts	<code>learning_rate</code> $\in [0.06, 0.24]$
generic ranges (from prior)		<code>n_estimators</code> $\in [150, 500]$, <code>subsample</code> $\in [0.65, 0.95]$, <code>min_child_samples</code> $\in [10, 60]$

Table 2: The fungal-prior mapping: mycorrhizal network topology measured on real data (Beiler 2010) constrains the LLM optimizer’s search space. Ranges are computed as center $\pm$ spread (Section 2.3).

reference evapotranspiration ET_0 (mm/day, FAO-56 Penman–Monteith), which scaled by crop coefficients determines how much water a grove must receive. A model that forecasts ET_0 from *yesterday’s* EO features enables proactive irrigation scheduling and provides the water-use traces that export certifications increasingly demand.

3.2 Data and features

We assembled 10 years (2016–2026) of daily observations for La Ligua (32.45°S, 71.23°W) from two independent open sources: Open-Meteo/ERA5 ($T_{\max}$, $T_{\min}$, precipitation, shortwave radiation, FAO-56 ET_0) and NASA-POWER/CERES (independent temperature, humidity, precipitation, solar radiation, pressure). After removing NASA-POWER fill values, 3,900 daily records remain (>94% coverage). We engineer 23 features: calendar harmonics, autoregressive lags of ET_0 (1–7 days), lags of radiation/temperature/humidity/precipitation, and 7-day rolling means. The target is ET_0 at $t+1$, with a strict temporal split: train < 2023 (2,543 days), validation 2023–2024 (731), test 2025 (364) — no leakage across the boundary.

3.3 Setup and methods

All methods share the same 9-dimensional LightGBM search space, data split, evaluation (validation RMSE) and budget of 12 trials: (i) **Random Search** samples uniformly; (ii) **Optuna TPE** (Akiba et al., 2019) is the classical BO baseline; (iii) **OPRO-vanilla** (Yang et al., 2024) prompts an LLM (Qwen-plus via DashScope) with the history over the *full* unconstrained space; (iv) **Myco-Opt** prompts the *same* LLM with the history *and* the mycorrhizal-constrained ranges of Table 2. Three seeds per method; LLM temperature 0.7 with strict JSON parsing — failed parses fall back to a uniform draw, making the comparison more conservative.

4 Results

Table 3 reports validation and test RMSE averaged over 3 seeds; Figure 1 shows the convergence curves (best validation RMSE as a function of the evaluation budget).

The prior is what makes the LLM optimizer work. At the *first* evaluation (the pure prior, before feedback), OPRO-vanilla’s proposal has validation RMSE 0.988 across seeds, whereas Myco-Opt’s first proposal averages 0.616 (0.598 best seed). After 12 trials, unconstrained OPRO converges to 0.621 — *still worse than Myco-Opt’s very first proposal*. The prior compresses the search space into the statistically plausible region, exactly what a scarce-budget regime needs.

Classical TPE remains the strongest method at this budget (test RMSE 0.606 vs. 0.614) — when each evaluation costs under a second, TPE’s surrogate amortizes quickly; the regime where Myco-Opt’s advantage is structural is the *expensive-evaluation* regime of neural EO pipelines (Section 5).

Method	Best val RMSE	Test RMSE	Test R^2
Optuna TPE	0.594	0.606 $\pm$ 0.002	0.807
Random Search	0.596	0.606 $\pm$ 0.003	0.807
Myco-Opt (prior)	0.604	0.614 $\pm$ 0.003	0.802
OPRO-vanilla (no prior)	0.621	0.627 $\pm$ 0.005	0.793

Table 3: HPO results on the ET_0 forecasting task, budget = 12 evaluations, 3 seeds. Test set: 2025 (364 days). Myco-Opt improves the LLM optimizer by 0.013 RMSE (2.1%) over OPRO-vanilla at the same budget and with the same LLM.

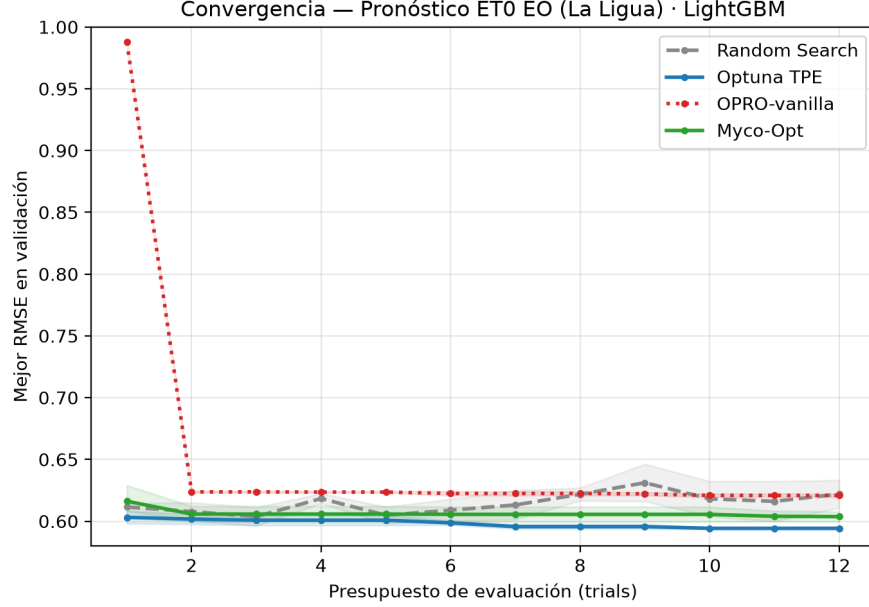


Figure 1: Convergence of the four methods on validation (mean over 3 seeds, shaded = ± 1 std). The unconstrained LLM (OPRO-vanilla) starts poorly and plateaus high; the mycorrhizal prior (Myco-Opt) starts competitive and converges faster. Classical TPE remains the strongest baseline at this budget.

5 Discussion and limitations

OPRO-vanilla and Myco-Opt differ only in the prior constraining proposal ranges; the consistent improvement (test RMSE 0.627 $\rightarrow$ 0.614; first-trial 0.988 $\rightarrow$ 0.616) shows that *quantitative biological topology transfers* into useful inductive bias for LLM-based HPO. Limitations: classical BO still wins at cheap budgets (the prior’s advantage grows with evaluation cost); we use one network and one EO task (follow-up will use GlobalFungi (Větrovský et al., 2020) and neural EO models); the topology-to-hyperparameter mapping is interpretable but partly heuristic; and we use only measured statistics, acknowledging the critique of common-mycorrhizal-network inference (Karst et al., 2023). Aligned with this workshop’s open-infrastructure emphasis, all code, data and the prior will be released open source.

6 Conclusion and future work

We presented Myco-Opt, an LLM-driven HPO optimizer whose search space is constrained by metrics measured on a real mycorrhizal network. On an agricultural EO benchmark (daily ET_0 forecasting for avocado orchards in drought-stricken Chile), the prior turned an unconstrained

LLM that plateaus at RMSE 0.627 into one reaching 0.614 at the same budget — with a first proposal (0.616) beating the unconstrained LLM’s converged result. Classical TPE remains the reference at cheap budgets; Myco-Opt’s promise lies in expensive-evaluation regimes, to be validated with neural pipelines. We aim to seed discussion of *biological inductive biases for AutoML in EO*.

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